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Pavement distress detection and classification based on YOLO network

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ABSTRACT
The detection and classification of pavement distress (PD) play a critical role in pavement maintenance and rehabilitation. Research on PD automation detection and measurement has been actively conducted. However, types of PD are more necessary for road managers to take effective actions. Also, lack of a unified PD dataset leads to absence of a benchmark on various methods. This study makes three contributions to address these issues. Firstly, a large-scale PD dataset is prepared. This dataset is composed of 45,788 images captured with a high-resolution industrial camera installed on vehicles, in a variety of weather and illuminance conditions. Each image is annotated with bounding box representing location and type of distress. Secondly, a deep learning-based object detection framework, the YOLO network, is adopted to predict possible distress location and category. Comprehensive detection accuracy reaches 73.64%. The processing speed reaches 0.0347s/pic, as 9 times faster than Faster R-CNN and only 70% of SSD. Finally, the applicability of model under various illumination conditions is also explored. The results reveal that the method significantly outperforms with appropriate illumination. We conclude that the proposed YOLO-based approach is able to detect PD with high accuracy, which requires no manual feature extraction and calculation during detecting.

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Pavement distress; object detection; image classification; YOLO network

Introduction

Pavement distress (PD) detection plays a pivotal role in road maintenance and rehabilitation. The occurrence of PD can result in a reduction in driving comfort and driving safety, besides, the lack of timely repair can also cause a reduction in road life. Therefore, early wide range detection and disposal of PD can bring considerable social and economic benefits. In the past decades, there is a lot of progress in both research of PD detection system and algorithm on PD identification.

Research of pavement distress detection system

The development of PD detection techniques have gone through three stages, from early traditional manual inspection to semi-automatic inspection at the end of the twentieth century, to the current non-destructive automatic detection.

The earliest adopted PD detection technique was manual inspection. It was necessary to draw distress location map and record the specific features such as the length, orientation and severity of PD. Since the labour-based inspection depended on the experience and knowledge of the test personnel, the test results were susceptible to human factors. And given its unexpected perception errors and low detection efficiency, it is not suitable for application in the wide range detection of PD. Meanwhile, in most situations, onsite inspections required road closure to implement. It might also impact traffic flow.

Gradually, PD detection techniques based on semi-automatic inspection became mainstream. The GERPHO system developed by the LCPC Road Management Department of France was developed and put into use. The high-speed camera and vehicle positioning system realized the synchronous acquisition of road surface images (Er-yong 2009). It greatly reduced the impact on traffic flow. However, there were shortcomings such as large post-processing workload, long time-consuming, and single detection function. With the development of video and tape storage technology, a rapid detection system based on video appeared in the 1980s. The most representative one is the Komatsu system in Japan. Cameras on two sides of the vehicle illuminated the road surface and obtained road surface images. The collected data was transmitted to the storage device based on the signal processor and sensor (Kim 2008). After the mid-1990s, the rapid development of CCD digital imaging and computer vision processing promoted the rapid detection system with low-cost, high-resolution, high-acquisition digital acquisition devices. The grayscale or colour of the object image was directly converted in the form of a pixel matrix with digital cameras, without the need of analogy/digital conversion, and its spatial resolution and image acquisition rate were much higher than analogy cameras. The most successful commercial products of this period were the ARAN system developed by ROADWARE in Canada, the Hawkeye 2000 system developed by ARRB in Australia, and the PAVUE system developed by Sweden IME. However, it was necessary to increase the artificial illumination source to obtain a better

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shooting effect (Er-yong 2009). Since 2000s, breakthroughs were made in the detection based on high-speed line scanning digital camera and laser. The main technical features were the use of photographic and infrared laser illumination technology to make image quality more stable. Representative of this type of detection system is the LRIS system produced by Canada's INO Corporation and the multi-function road condition detection system produced by the US ICC company(Jian 2017). Besides, the PD detection based on infrared thermal imaging technology used temperature difference of road surface and PD to reflect damage. The image processing algorithm was correspondingly simple. However, since the infrared equipment is costly and is susceptible to temperature and humidity, it had not been promoted widely(Du et al. 2017). The above techniques can capture road surface conditions and reproduce PD through the indoor interpretation device, while the technicians confirm various diseases in the laboratory and manually input the results into the database. They also need manual participation.

Eventually, integrated non-destructive automatic inspection systems were gradually promoted in the market. The Road Traffic Intelligent Detection and Equipment Engineering Technology Research Centre of Chang'an University successfully developed the CT-501A high-speed laser road inspection vehicle(Jian-feng 2010). The detection system used an area array camera to collect road images combined with laser, which can realize high-precision and rapid detection of road surface flatness, rutting, construction depth and damage. The quality of the road image was good enough to provide a basis for the automatic identification and classification of PD. Zhonggong Gaoke Maintenance Technology Co., Ltd. developed the road detection vehicle CiCS. It used line array camera to collect road surface image, combined with structured light illumination. It can achieve a detection width of 3.6 m in the horizontal direction, the top speed of 100 km/h and a highest detection accuracy up to 1 mm,. The company's cracking image analysis system, CiAS (Cracking Image Analysis System), automatically handled road critical indicators including cracks, ruts and flatness(Jian-feng 2010). The 3D laser scanning was used to obtain the fine contour of the road surface. A laser line was projected on the near-flat pavement and the camera detected the shape of the line under an angle which reflected the depth of the surface(Coenen and Golroo 2017). The advantage of this technology was that it acquired the electric cloud data of the road surface, without complex texture and colour information, and brought convenience to the later image processing. The current problem of the system was that the amount of detected data was large, and the reliability still needed to be further improved. Representative systems were the DHDV test vehicle developed by the University of Arkansas, Wang *et al.*(Wei-guo 2005), the CrackScope system developed by the University of Texas at Austin (Bu-gao 2006).

Research of algorithm on pavement distress identification

The automation detection of PD also required robust algorithms of high level of intelligence. Vastly automated analysis tools for PD detection were vastly proposed by private and public endeavours on worldwide basis over the past decades. With human's efforts in developing mathematical models simulating cognition capabilities, these methods were steadily improved.

Traditional automation methods on PD detection were diverse. Thresholding algorithms were proposed to find cracks by setting global or local thresholds (Oliveira and Lobato Correia 2009), under the assumption that cracks can be identified as local intensity minima. Segmentation-based methods were proposed to enable detection to be conducted at the block level (Ying and Salari 2010). Edge detectors were widely used to detect the edges of pavement cracks (Nisanth and Mathew), though with incapability to detect complete crack profiles. Filter-based algorithms were developed to find cracks of strong responses to predesigned filters (Zalama et al. 2014). Wavelet Transform was applied to decompose the original data into different frequency sub bands, given the assumption that cracks are primarily preserved in high frequency sub bands (Wang et al. 2007). However, there were also some limitations for those methods: 1) requiring high process ability and complicated operation (Cao et al. 2014); 2) being easily influenced by light and shadow (Li et al. 2010); 3) and with incapability in detecting complete PD profiles. For example, Cao *et al.* (Cao et al. 2014) adopted a median conversion method to complete the automatic recognition and feature measurement of pavement crack distress, but it was difficult for the computation to achieve rapid batch detection. Li *et al.* (Li et al. 2010) applied the least-cost path search algorithm for crack detection. The algorithm had high accuracy, but was susceptible to light. Li *et al.* (Li et al. 2011) developed the F*Seed-growing algorithm to extract the pavement features. This method assumed that the darker parts of the pavement were cracks while such results could be invalid with shadows and some of the cracks were automatically ignored.

Machine learning provided new ideas for PD detection. Many successful applications of machine learning techniques were reported in the field of transportation engineering, such as traffic incident detection (Samant and Adeli 2000), work zone capacity estimation (Jiang and Adeli 2003), traffic flow forecasting (Jiang and Adeli 2005), and traffic sign classification (Cireşan et al. 2012). There were pioneering applications of machine learning techniques, such as Artificial Neural Network (ANN) and Support Vector Machine (SVM), in classifying cracks on pavement surfaces (Daniel and Preeja 2014). However, these studies generally represented only one or two layers of features and cannot fully reflect the complexity of pavement surface. The more accurately the features were extracted, the better detection results would be. Different types of classifiers for patches of the image were applied to extract distress features. For example, the classifier was applied to extract Histogram of Oriented Gradient (HOG) features (Kapela et al. 2015) or Local Binary Patterns (LBP) (Varadharajan et al. 2014, Quintana et al. 2016). Shen *et al.* (Shen et al. 2013) used support vector machines (SVM) to identify the cross-scale distress of pavement image. Its central idea was to apply the principle of structural risk minimization to the field of classification. Zakeri *et al.* (Zakeri et al. 2013) described an approach that used a multi-layer perceptron network in combination with frequency features and image histograms. The algorithm improved the accuracy and efficiency of PD image recognition, which achieved quickly and accurately the identification of the degree and extent of PD. However, the process of extracting features often demanded a lot of manpower and it could not distinguish the types of PD. Therefore, it called for more effective methods for PD detection and classification.

Deep Learning were put forward to solve those problems, with the expansion of sample data scale, the improvement of computing performance (especially with GPUs widely used in large-scale parallel computing) and algorithm model innovation. It can learn from experience and understand complex problems based on a hierarchy of concepts (Bengio et al. 2016) that were defined and learned through increasing levels of abstraction (Murphy 2012). Particularly, deep Convolutional Neural Networks (CNN) demonstrated success in large-scale object recognition problems without requiring a separate feature extraction (Cha et al. 2017, Ren et al. 2017). CNN was a neural network structure proposed by Lecun *et al.* (Cun et al. 1989, Lecun 2015, LeCun et al. 2015), which was a highly non-linear mapping that can output the target features from the input image in a specific form. At the same time, CNN had a high degree of robustness to the scaling and tilt of the target geometry, which was sufficient to cope with complex forms of cracks, potholes, and other distress (Ijjina and Chalavadi 2016). In addition, CNN had strong back-to-back noise filtering capabilities (Jia et al. 2016, Mansanet et al. 2016), which was sufficient to overcome the spot noise and low comparability issues in the pavement inspection images. Tong *et al* (Tong et al. 2017a) used a CNN network to automatically identify cracks in asphalt pavements. The results showed that convolutional neural networks had better stability than other traditional detection algorithms. Eisenbach *et al.* (Eisenbach et al. 2017) automated this process to a high degree by applying deep neural networks. Tong *et al.* (Tong et al. 2017b) used the CNN network to identify, locate, measure, and reconstruct 3D reflections of Rydal images on the asphalt pavement. Du and Liu *et al.* (Du et al. 2019) proposed a dynamic method to estimate pavement friction level using computer vision. However, there were very few studies that directly used pavement images as input samples or use CNN to analyze PD. Besides, there was also a lack of intelligent means to directly select the exact class and location of the distress.

Research of YOLO network on transportation engineering

YOLO (You Only Look Once) is one of the real-time deep CNN methods that aims at detecting objects from images (Redmon et al. 2015). Using the CNN implemented in the YOLO platform (Redmon and Farhadi 2018), objects can be tracked, detected ('seen'), and classified ('comprehended') (Radovic et al. 2017). Many successful applications of YOLO network are reported in the field of transportation engineering. For example, YOLO has strong robustness and can complete vehicle detection tasks quickly (Yu-ning et al. 2016). YOLO as the Road Lane Detector is used to detect road tracks from video's frames and to provide additional information that can be helpful for the decision-making processes of the self-driving car (Nugraha et al. 2017). Pedestrian detection and tracking can also be realized based on YOLO network (Gao et al. 2018). YOLO network is widely applied in traffic management, rather than infrastructure management. PD detection is a critical issue in infrastructure management task, requiring further exploration. YOLO network can be applied to object detection of PD.

To fill the gap of research, we propose a new approach to detect and classify PD based on the YOLO network. The method directly captures pavement images as inputs, regardless of the interference from the environment. The images are model input, while output is a set of predicted frames including possible distress location and category. 10,000 images are set for training and 3,000 images for testing. Thanks to the unique characteristics of YOLO, image classification and object detection can be realized together. Hence, after labelling a large amount of distress marks for training and testing, the PD can be quickly detected and classified.

The remainder of this paper is organized as follows. Section 2 introduces PD detection and classification based on the YOLO network. Section 3 shows the data preparation process and verifies the proposed detection model through training and testing, and Section 4 investigates the feasibility of the method and discusses the results in different environmental conditions. Section 5 summarizes the study outcomes and concludes the paper.

Pavement distress detection and classification based on YOLO network

YOLO frame process Object Detection as a regression problem to spatially separate bounding boxes and associate class probabilities (Redmon et al. 2015). A single neural network is used to predict the coordinates of the bounding box directly from an entire image. The confidence of the object in bounding boxes and the probabilities of the object class are captured directly from full images in one evaluation. Since the whole detection pipeline is a single network, it can be optimized end-to-end directly on detection performance.

Firstly, the unified architecture of YOLO is extremely fast. Our base YOLO version 3 model processes images in real-time at 45 frames per second on the Titan X GPU. In order to meet the needs of road damage detection, it is necessary to collect a large amount of image data on the road surface. As calculation, the storage needs approximately 300pic/km*0.6MB/pic = 0.18GB/km. YOLO is faster among other deep learning frameworks that can save plenty of time.

Secondly, the YOLO is very robust to such near distance targets or small targets. Feature semantic information of low-level is relatively small, but the target location information is accurate; feature semantic information of high-level is rich, but the target location information is relatively rough. YOLOv3 adds multi-scale recognition capabilities to combine both advantages. YOLO v3 uses (like FPN) up-sample and fusion methods, which combines three scales (13*13, 26*26 and 52*52) and independently detects them on multiple scales of fusion feature maps. This method is especially effective in detecting small target objects and is suitable for the detection of pavement distress.

Finally, YOLO's framework is easy to deploy on the mobile side, so it can be used in mobile car devices or mobile phones to improve the intelligence of pavement distress detection.

Overall network architecture

The overall network architecture is shown in Figure 1.

The first layer is the input layer of multi-scale pixel resolutions, where each dimension indicates height, width, and channel (red, green, and blue), respectively. Input data pass through the architecture. The model divides the input image into an S × S grid. If the centre of an object falls into a grid cell, that grid cell is responsible for detecting that object.

Each grid cell predicts B bounding boxes. Each bounding box consists of 5 predictions: x, y, w, h and confidence. The (x, y) coordinates represent the center of the box relative to the bounds of the grid cell. The w *and* h are predicted relative to the whole image. The confidence scores reflect how confident the model is that the box contains an object and also how accurate it predicts. The confidence is defined as $Pr(Object)*IOU^{truth}_{pred}$. If no object exists in that cell, the confidence is zero. Otherwise the confidence score equals the intersection over union (IOU) between the predicted box and the ground truth.

Each grid cell also predicts C conditional class probabilities, $Pr(Class_i|Object)$. These probabilities are conditioned on the grid cell containing an object. It predicts one set of class probabilities per grid cell, regardless of the number of boxes. For evaluating YOLOv3 on PD, we use S = 7, B = 9, C = 7.

At test time, we multiply the conditional class probabilities and the individual box confidence predictions,

$$\begin{aligned} &Pr(Class_i|Object)*Pr(Object)*IOU^{truth}_{pred} \\ &= Pr(Class_i)*IOU^{truth}_{pred} \end{aligned} \tag{1}$$

which gives us class-specific confidence scores for each box. The scores encode both the probability of that class appearing in the box and how well the predicted box fits the object.

Convolution layer

A convolution layer performs the following three operations throughout an input array. First, it performs element-by-element multiplications between a sub-array of an input array and a receptive field. The receptive field is often called the filter, or kernel. The initial weight values of a receptive field are typically randomly generated. Bias can be set in many ways in accordance with networks' configurations. Both values are tuned in training using a stochastic gradient descent (SGD) algorithm. The size of a sub-array is always equal to a receptive field which is always smaller than the input array. Second, the multiplied values are summed, and bias is added to the sum.

An additional hyper-parameter of the layer is the stride. The stride defines how many of the receptive field's columns and rows (pixels) slide at a time across the input array's width and height. A larger stride size leads to fewer receptive field applications and a smaller output size, which also reduces computational cost, though it may also lose features of the input data.

Feature extraction

Feature extraction is performed by Darknet-53. From the 0th to the 74th layer, there are a total of 53 convolutional layers, and the rest are res layers. These convolutional layers are obtained by integrating convolutional layers with better performance from various mainstream network structures.

The res layers are derived from Resnet. Inputs and outputs are generally consistent, and the difference is calculating. In order to solve the phenomenon of gradient dispersion or gradient explosion of the network, it is proposed to change the layer-by-layer training of the deep neural network into phase-by-

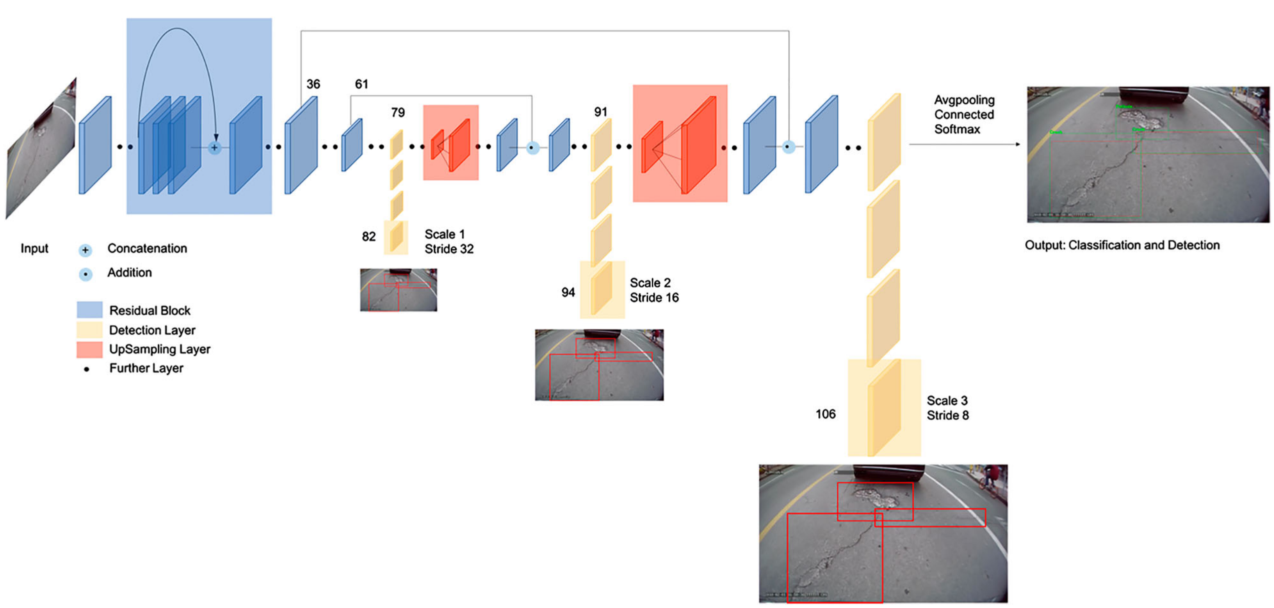


Figure 1. The YOLOv3 network Architecture.

phase training. It divides the DNN into several sub-segments, each of which contains shallow network layers. Then the shortcut connection method is used to make each small segment train the residual. Each small segment learns a part of the total loss, and finally makes the overall loss smaller.

The 75–105 layer is the characteristic interaction layer of the YOLOv3 network, which is divided into three scales to obtain features. In each scale, the convolution kernel (3×3 and 1×1) implements local features between the feature maps interactions and the fully-connected layer implements global feature interactions. The architecture of Darknet-53 is as follows (Figure 2):

Activation function

A logistic activation function for the final layer is used as sigmoid function:

$$sigmoid(x) = \frac{1}{1+e^{-x}} \tag{2.1}$$

And all other layers use the following leaky rectified linear activation:

$$\Phi(x) = \begin{cases} x, & if\ x > 0 \\ 0.1x, & otherwise \end{cases} \tag{2.2}$$

	Type	Filters	Size	Output
	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
1×	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
	Convolutional	128	3 × 3 / 2	64 × 64
2×	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
	Convolutional	256	3 × 3 / 2	32 × 32
8×	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
	Convolutional	512	3 × 3 / 2	16 × 16
8×	Convolutional	256	1 × 1	
	Convolutional	512	3 × 3	
	Residual			16 × 16
	Convolutional	1024	3 × 3 / 2	8 × 8
4×	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

Figure 2. Architecture of Darknet-53 (revised based on Joseph Redmon *et al*).

Loss function

During training, we use sum of squared error loss. The loss function is as follows:

$$\begin{aligned} loss = {} & \lambda_{coord}\sum_{i=0}^{S^2}\sum_{j=0}^{B}\mathbb{1}_{ij}^{obj}[(x_i-\hat{x_i})^2+(y_i-\hat{y_i})^2] \\ & +\lambda_{coord}\sum_{i=0}^{S^2}\sum_{j=0}^{B}\mathbb{1}_{ij}^{obj}[(\sqrt{\omega_i}-\sqrt{\hat{\omega_i}})^2+(\sqrt{h_i}-\sqrt{\hat{h_i}})^2] \\ & +\sum_{i=0}^{S^2}\sum_{j=0}^{B}\mathbb{1}_{ij}^{obj}(c_i-\hat{c_i})^2+\lambda_{noobj}\sum_{i=0}^{S^2}\sum_{j=0}^{B}\mathbb{1}_{ij}^{noobj}(c_i-\hat{c_i})^2 \\ & +\sum_{i=0}^{S^2}\mathbb{1}_{i}^{obj}\sum_{c\in classes}(p_i(c)-\hat{p_i}(c))^2 \end{aligned} \tag{3}$$

where $\mathbb{1}_i^{obj}$ denotes if object appears in cell i and $\mathbb{1}_{ij}^{obj}$ denotes that the jth bounding box predictor in cell i is responsible for that prediction. $\hat{x}, \hat{y}, \hat{\omega}, \hat{h}, \hat{C}, \hat{p}$ are predicted value, while (x, y, ω, h, C, p) are labeled value.

We optimize for sum-squared error in the output of our model. YOLOv3 predicts multiple bounding boxes per grid cell. At training time, we only select one bounding box predictor to be responsible for each object, whose prediction has the highest current IOU with the ground truth. This leads to specialization between the bounding box predictors. Each predictor gets better at predicting certain sizes, aspect ratios, or classes of objects, improving overall recall.

Bounding box prediction

The network predicts 4 coordinates for each bounding box, t_x, t_y, t_w, t_h. If the cell is offset from the top left corner of the image by (c_x, c_y) and the bounding box prior has width and height p_w, p_h, then the predictions correspond to (Figure 3):

$$\begin{aligned} b_x &= \sigma(t_x) + c_x \\ b_y &= \sigma(t_y) + c_y \\ b_w &= p_w e^{t_w} \\ b_h &= p_h e^{t_h} \end{aligned} \tag{4}$$

YOLOv3 predicts an objective score for each bounding box using logistic regression. This should be 1 if the bounding

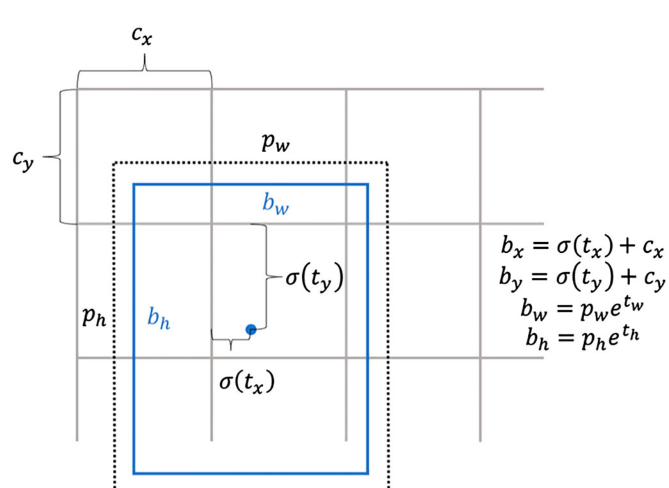


Figure 3. Bounding boxes with dimension priors and location prediction (revised based on Joseph Redmon *et al*).

box prior overlaps a ground truth object by more than any other bounding box prior. If the bounding box prior is not the best but does overlap a ground truth object by more than some threshold, we ignore the prediction. The threshold of 0.5 is used. The network only assigns one bounding box prior for each ground truth object.

Experimental tests and results

Data preparation

The image dataset used in this paper for object detection is composed of 45,788 PD images captured with a high resolution industrial camera installed on a vehicle, with 59366 instances of PD included in these images, covering almost 200 km of urban trunk roads (Du 2018). The details of the actual implementation are as follows: 1) choose asphalt roads; 2) try to avoid traffic congestion; 3) keep the speed not exceeding 80 km/h; 4) try to minimize lane change operations.

The complexity of PD detection lies in the diversity of road surface conditions. The diversity of road surface illumination shadows greatly affects PD detection. The dataset includes a variety of conditions with different illumination and roadside object shadows, which can meet the accuracy of the training algorithm in various scenarios in the future.

Data annotation

After being obtained by the mentioned data collection method, the extracted sample images (more than 30,000 images) are annotated by the labelling method of the labelImg-master_v1.5.2 version. The PD categories are classified into seven classes, with the number of each as follows: 'Crack (longitudinal cracking& transverse cracking)': 14505, 'Pothole': 2163, 'Net (alligator cracking)': 4710, 'Patch-Crack (patched longitudinal cracking& patched transverse cracking)': 18931, 'Patch-Pothole': 6518, 'Patch-Net (patched alligator cracking)': 1358, 'Manhole': 11181. The data labelling in training-based method is actually also a labour-intensive procedure. Yet, it is also worth mentioning that the proposed method replace huge labor cost when identifying distress by relatively small one in algorithm pre-preparation period. Besides that, data labelling can be tested by cross-checking, which can greatly improve the accuracy.

Data pre-processing

In view of the unbalanced data quantity of various PD types, in order to prevent under-fitting during the training of the deep learning network, it is often necessary to supplement the amount of data. We use the image data generator in Keras frame to amplify the original image. It can easily implement various forms of image amplification methods, such as rotation, horizontal and vertical panning, horizontal and vertical flipping, and zooming on a certain part. The figure below shows an image that is randomly rotated, partially enlarged, etc. (Figure 4):

Before training, it is necessary to pre-process the image and normalize the input data. If the range of the input data differs too much, it may affect the subsequent learning process. The image data should be changed to zero-centred. Because the training rate of gradient descent algorithm used for optimization may be affected due to the constantly positive or negative input. Besides, due to the different scales in two dimensions, the convergence rate of the model may also be affected.

The dataset is composed of RGB images, so each image is imported in the form of a three-dimensional matrix with each value in the matrix corresponding to the pixel at its position in the image. We adjust the distribution of the original

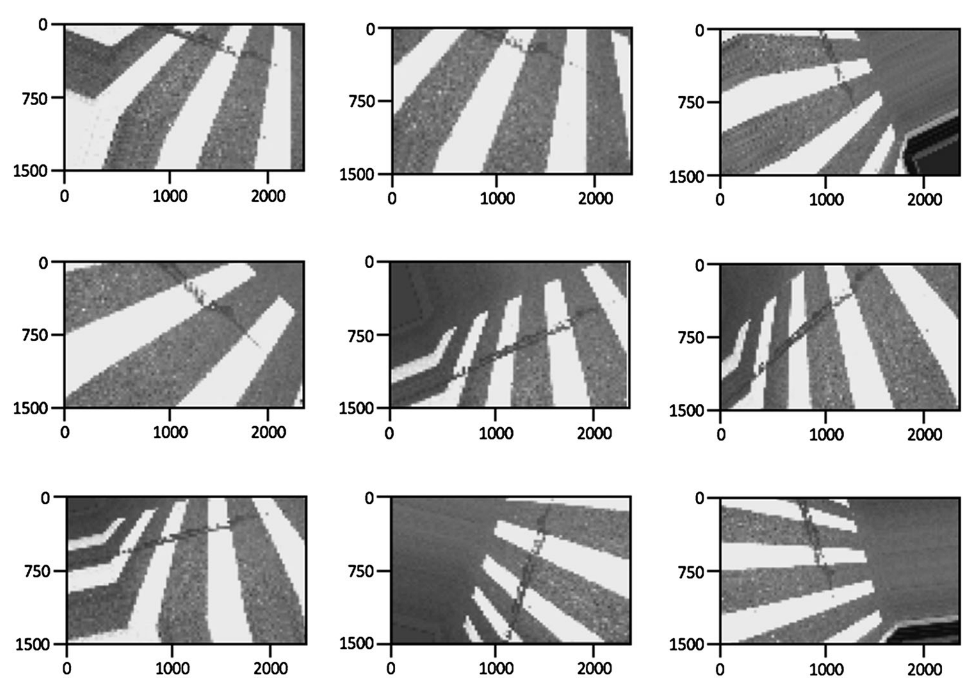


Figure 4. Examples of data amplification.

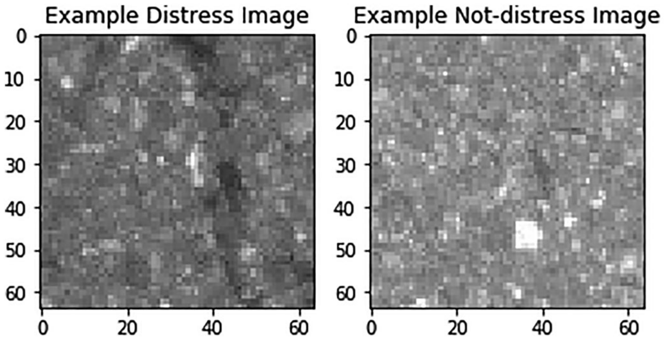


Figure 5. Example of the distress and not-distress classes.

image to zero-centred. Suppose the input picture size is $W \times H \times 3$, the total number of samples is M, the current number of the sample is i, and the matrix of the input picture is set to X. Then the zero-centred pixel matrix is:

$$X_{mean} = \frac{1}{M}\sum_{i=1}^{M} X_i \qquad (5)$$
$$X_{i,0} = X_i - X_{mean}$$

The image data is further standardized through dividing the pixel value in each image matrix by the standard deviation of pixel value on the corresponding point from all input images.

$$X_{mean,0} = \frac{1}{M}\sum_{i=1}^{M} X_{i,0}$$
$$X_{std} = \frac{1}{M}\sum_{i=1}^{M} (X_{i,0} - X_{mean,0})^2 \qquad (6)$$
$$X_{i,std} = X_{i,0}/X_{std}$$

The dataset is divided into 3 parts: training set, validation set and test set. The image quantity ratio of the 3 sets is roughly 7:3:3. The training set is used to train the model and determine the parameters, such as weights; the validation set is used for hyper-parameter optimization and selection; testing set is purely used to test the generalization ability of the trained model.

Binary classification

To understand the possibility of the algorithm beforehand, Binary Classification is trained and tested on the GAPs dataset (Eisenbach et al. 2017). The dataset includes a total of 1969 grey valued images (8 bit), partitioned into 1418 training images, 51 validation images, and 500 test images. The image resolution is 1920×1080 pixels with a per pixel resolution of 1.2 mm×1.2 mm. The images have been annotated manually by trained operators at a high-resolution scale. An actual distress is cut into small scale images with a size of 64×64 pixels. The 64×64 pixels scale images include 3804 distress images and 28196 not-distress images with separate labels. Here is an example of the distress and not-distress classes (Figure 5):

Training

We train on full images with no further mining. Operations such as multi-scale training, various data augmentation, batch normalization, etc. are used. The Darknet neural network framework is applied for training and testing.

All of the described training and testing tasks in this article are performed on a workstation with four GPUs (CPU: Intel Xeon E5-2650 v4 × 2ea @2.9 GHz, RAM: 128GB and GPU: Nvidia Geforce Titan X × 4ea).

We randomly select 3 sets of data as training and cross validation (7:3) samples, with the size of 10,000 images, 20,000 images, and 30,000 images, respectively. The loss curve of the training is concerned and its convergence is the basis for stopping training.

The parameters of the network are relatively numerous. Hence, training from the beginning may lead to the overfitting phenomenon. Therefore, we adopt the transfer learning method to train our model. The pre-trained weights are initialized by the weights from the model trained by VOC datasets. Because the negative samples (environment) presented by the images of the VOC datasets has the similar characteristics as PD. They belong to the same domain adaptation, and the method of extracting features is the same. The weight is further adjusted to the network in the training. The underlying network is frozen, and only the parameters of the upper layer network are adjusted through training. This is because the deep network is extracted at the bottom layer with often a relatively macroscopic feature, while upper network extracted with relatively subtle specific features. Multiple parameter combinations are tried via parallel computing, the best one is selected through validation.

Testing

The sample quantity of PD types refers to the sample numbers of seven PD types. The average precision (AP) value is the area

Table 1. Test results comparison of different types of PD on YOLOv3 network.

Class	10,000 images			20,000 images			30,000 images		
	Sample Quantity of PD types	AP	F1 score	Sample Quantity of PD types	AP	F1 score	Sample Quantity of PD types	AP	F1 score
Crack	3016	48.0%	0.4381	4842	61.0%	0.6390	7330	74.0%	0.7684
Patch-Crack	2071	67.2%	0.5511	6870	79.2%	0.7870	7650	80.8%	0.8438
Pothole	426	31.8%	0.3631	721	60.1%	0.6108	1093	60.2%	0.6703
Patch-Pothole	924	53.3%	0.4685	1849	64.3%	0.6855	3294	75.3%	0.8025
Net	635	42.2%	0.4107	1205	59.2%	0.6273	2356	63.1%	0.7101
Patch-Net	343	43.5%	0.5123	622	41.2%	0.4882	686	69.4%	0.6905
Manhole	2885	79.4%	0.8215	5203	82.7%	0.8494	9567	92.7%	0.9320

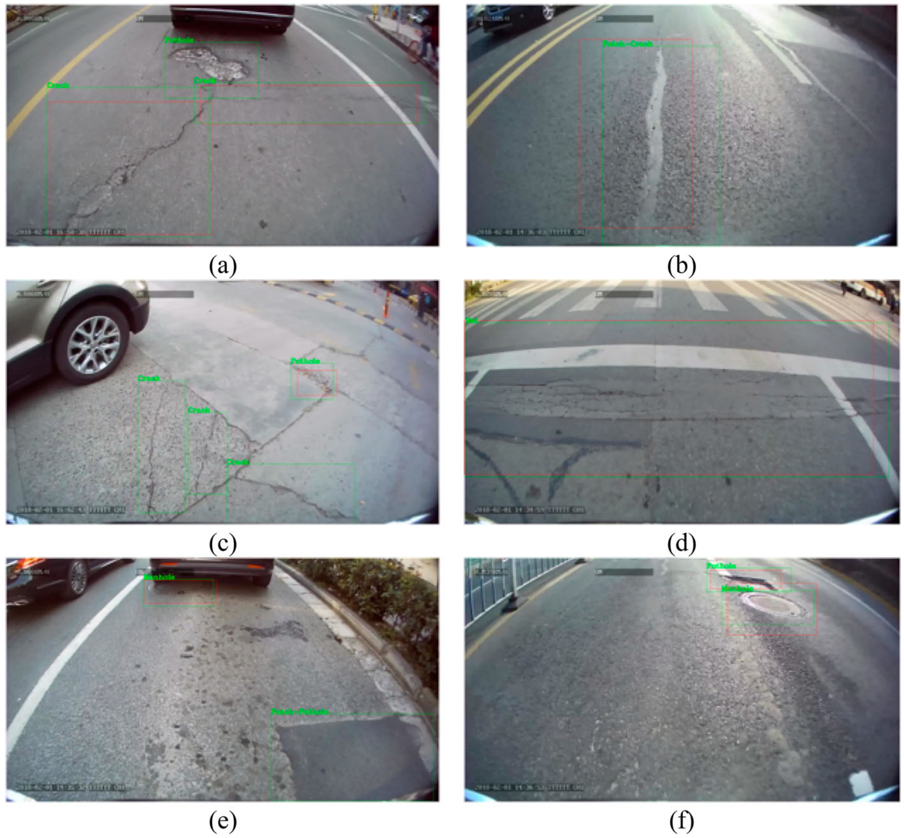


Figure 6. Successful detection. The red box in the picture is a sample label box, and the green box is the result of YOLOv3 network.

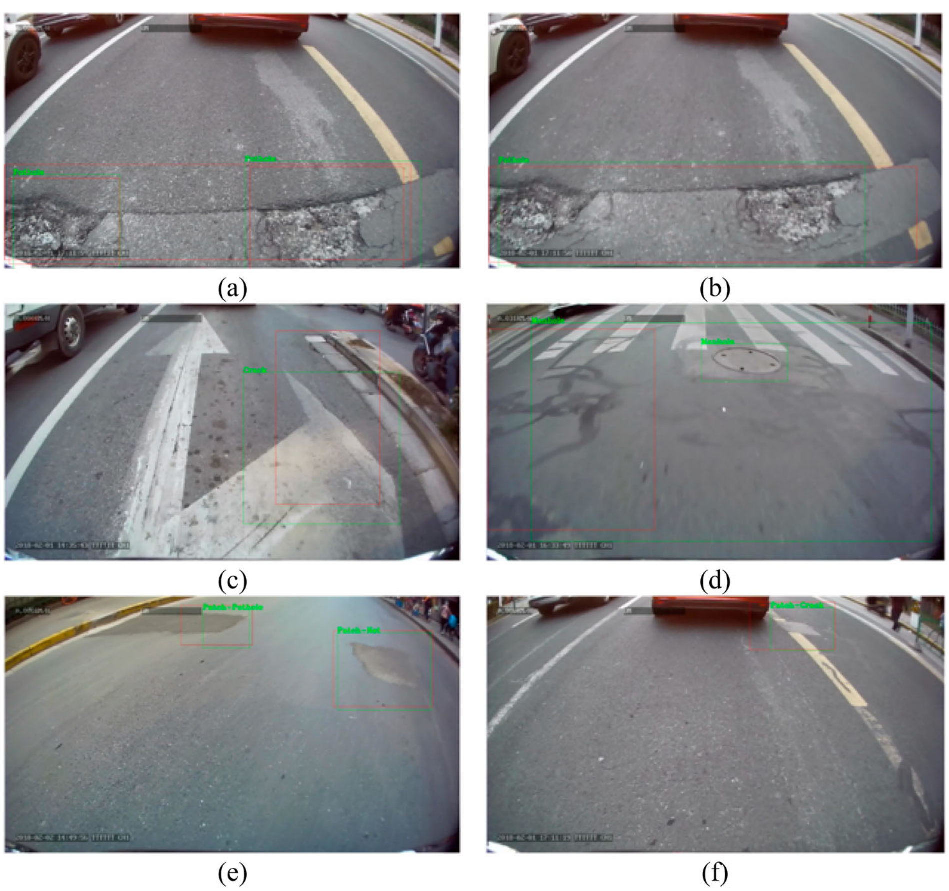


Figure 7. Unsuccessful detection. The red box in the picture is a sample label box, and the green box is the result of YOLOv3 network.

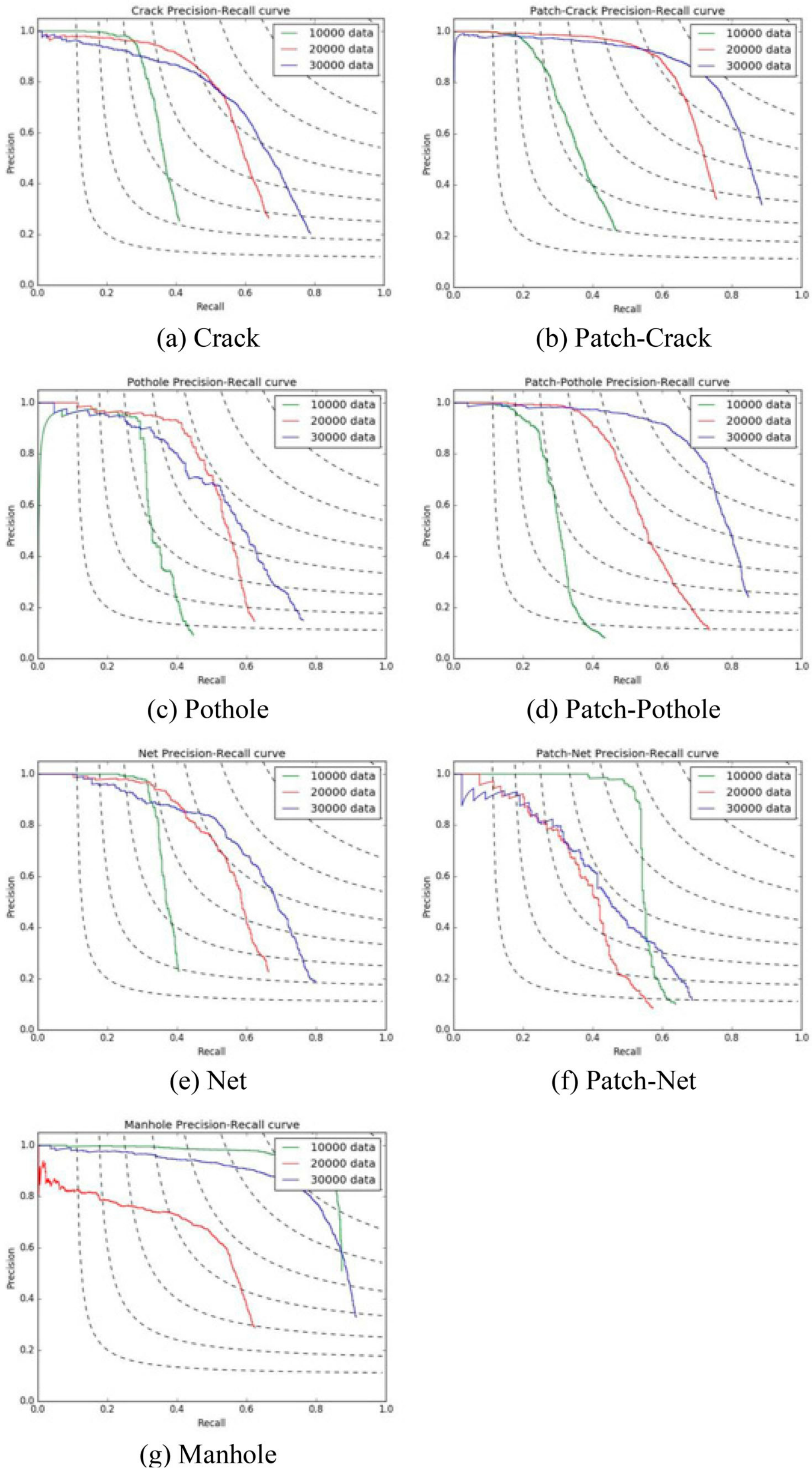


Figure 8. Precision-recall diagram on different size of training samples. Dashed lines show the contour plots of the derived performance measures F1 scores.

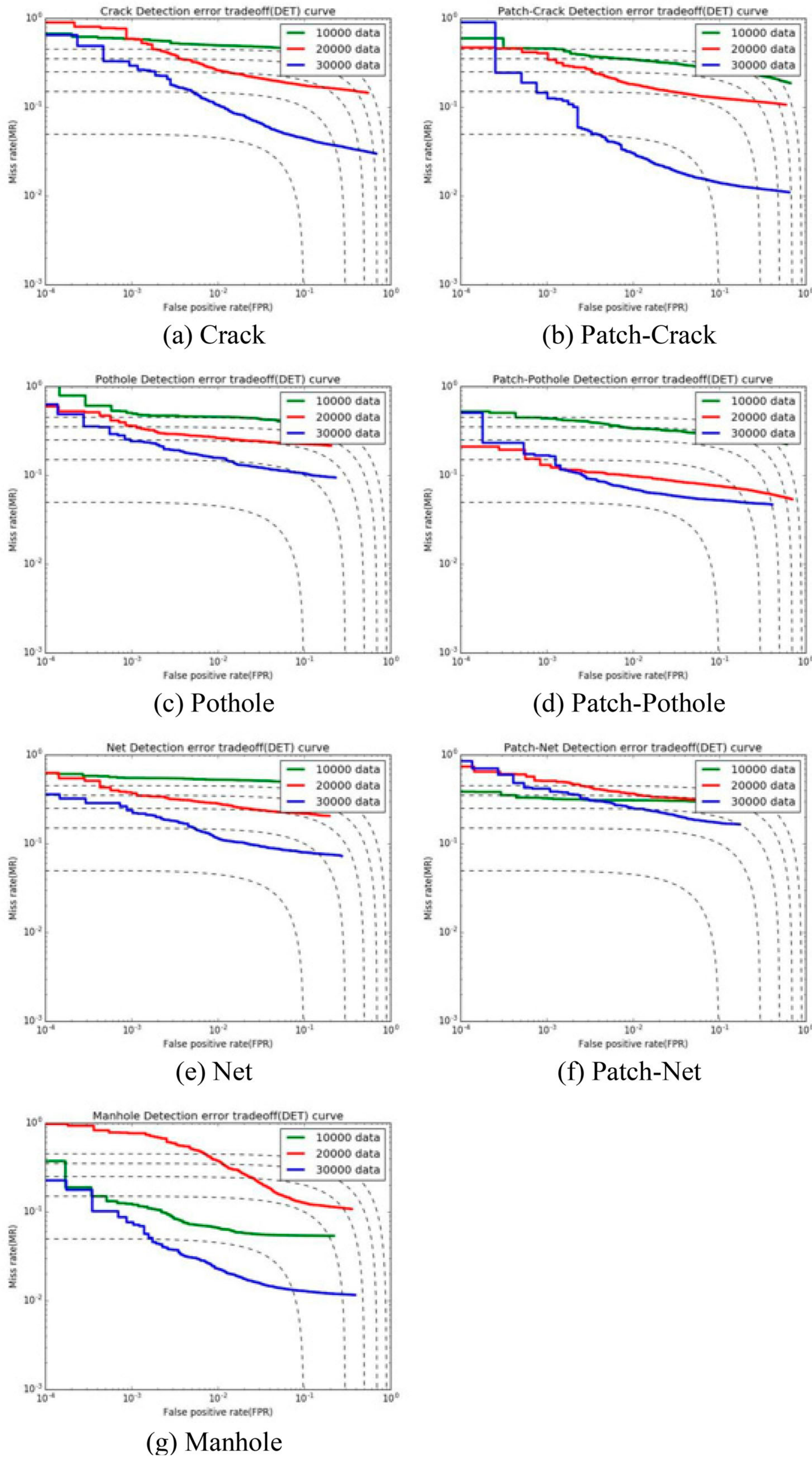


Figure 9. Detection error trade-off (DET) curve on different size of training samples. Dashed lines show the contour plots of the derived performance measures balanced error rate (BER).

Table 2. Comparison of different algorithms on AP/F1 score and operation time.

Types of PD	YOLOv3 (Batch = 64)		YOLOv3 (Batch = 96)		Faster R-CNN		SSD	
	AP	F1 score	AP	F1 score	AP	F1 score	AP	F1 score
Crack	48.5%	0.4375	45.8%	0.4567	49.3%	0.5488	35.7%	0.3467
Patch-Crack	69.8%	0.6673	69.2%	0.7126	72.3%	0.7658	56.0%	0.5721
Pothole	38.6%	0.3631	38.3%	0.4088	34.9%	0.3356	33.8%	0.3278
Patch-Pothole	63.3%	0.6855	61.6%	0.6579	62.7%	0.6753	55.1%	0.5689
Net	50.6%	0.5134	48.6%	0.5076	53.5%	0.5690	46.6%	0.4875
Patch-Net	41.8%	0.4289	40.9%	0.4055	41.9%	0.4432	34.4%	0.3567
Manhole	84.6%	0.8934	83.1%	0.8896	86.1%	0.9045	83.1%	0.8756
Operation Time/s	364.965		366.415		4359.300		496.696	

under the Precision-Recall curve. The specific discussion of Precision-Recall curve is in DISCUSSION section.

F score is often used to show the method performance, calculated as follows:

$$F(\beta)score = (1+\beta^2)*\frac{Precision*Recall}{\beta^2*Precision + Recall} \tag{7}$$

For the convenience and intuitiveness of the calculation, we use a simple average F1 score with the AP value to show the method performance on whole dataset. The testing results based on 3 scales of training samples are shown in Table 1.

Patched PD can be detected more easily than unpatched ones (Table 1). Patched PD are coated with a new repair material on the basis of original distress, which makes the surface characteristics of the road change greatly. Besides, manhole type has the best AP value, for its features differ mostly from pavement texture and roughness.

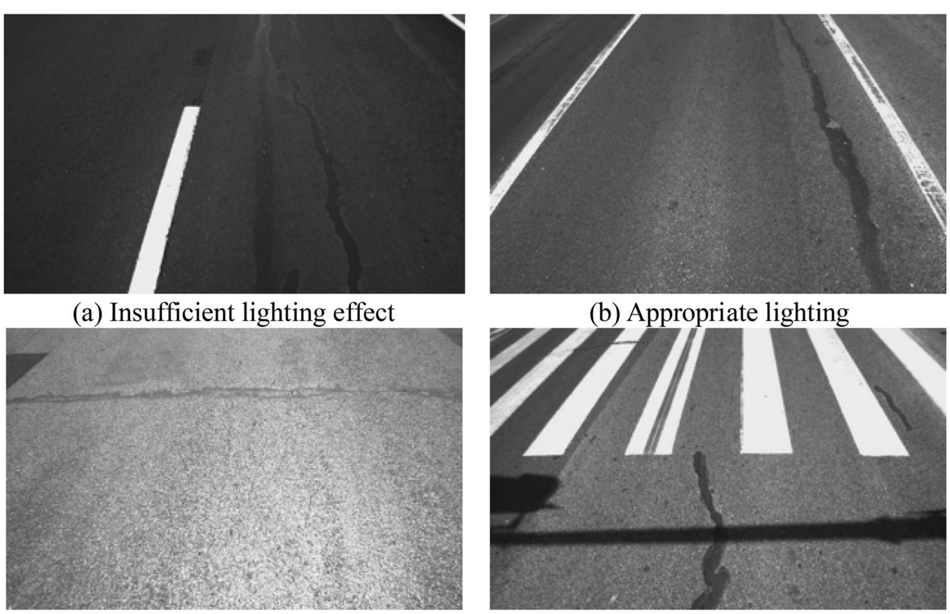
The visual outputs of successful and unsuccessful detection are shown in Figures 6 and 7.

Figure 7. (a), (b) have the same image as input, however, with different test results. This is because during the marking process, the labels of the same sample are different. Thus, YOLO encounters confusion about which samples should be based on, leading to a deviation in the process of learning. Figure 7. (c) has a tolerable deviation. It is mainly due to the fact that the characteristics of curbs are too similar to those of PD, resulting in errors of network judgement. There is also a deviation in Figure 7. (d) mainly because there is too much dirt in the whole area, coinciding with the features of a manhole. The network mistakenly identifies a large area as a manhole that is supposed to be small. Figure 7. (e) is wrongly detected because the label box is not complete. Figure 7. (f) has a wrong label due to the same reason.

7 main types of distresses on road surface are detected and classified based on our method, including crack, pothole, net, patched crack, patched pothole, patched net and manhole. They basically cover the most common distresses on road surface that affect road performance and driving quality.

Pavement engineers are more concerned about the types of distresses together with their severity and extent to guide maintenance more effectively. The bounding box drawn on the detecting results shows the length and width of the smallest circumscribed rectangle of the distress. Considering that the image can only obtain plane dimension information, only two-dimensional parameters are recorded to reflect the severity and extent of the distress. Combining the information of the depth dimension, the distress characteristics can be described more accurately, which requires further study.

Specific classification of longitudinal cracking and transverse cracking is processed further. We take the direction of



(a) Insufficient lighting effect (b) Appropriate lighting

(c) Strong lighting effect (d) Street light shadow effect

Figure 10. Examples of dataset with (a) Insufficient lighting effect, (b) Appropriate lighting, (c) Strong lighting effect and (d) Street light shadow effect.

Table 3. F1 score comparison of different types of PD on different image illuminance.

Types of PD	Insufficient light	Appropriate light	Strong light	Appropriate light with unexpected shadows
Crack	0.5603	0.8168	0.6125	0.6435
Patch-Crack	0.6043	0.8078	0.7019	0.6565
Pothole	0.5021	0.7161	0.6027	0.5449
Patch-Pothole	0.5778	0.7838	0.7450	0.7432
Net	0.5534	0.8235	0.6753	0.6944
Patch-Net	0.5028	0.7742	0.6924	0.6569
Manhole	0.6630	0.8534	0.6867	0.7046

the image parallel to the lane line as the H direction, the direction perpendicular to the lane line as the W direction. The size of the bounding box in the H direction is set as 'y', and the size in the W direction is set as 'x'. Therefore, the distresses with y > x are classified as 'longitudinal cracking' while the ones with x > y is classified as ' transverse cracking'.

Discussion

Potential of accuracy improvement

It is proved that the performance of the network has improved steadily with the increase in the size of training samples. For a visualization of the performance, curves in the precision recall (PR) diagram, and the detection error trade-off (DET) diagram that are derived from the ROC curve on different size of training samples are shown in Figures 8 and 9. The blue lines are the final results of precision-recall diagram on training samples with 30,000 images. The area under the curves (AP value) increases as the size of training samples grows (from green line, red line to blue line) in Figure 8**,** while it decreases in Figure 9. This is consistent with common sense. With the increasing size of samples for training, the performance of the network improves, the robustness of the network on PD becomes stronger; the accuracy, higher; and the error rate, lower.

Algorithm comparison on accuracy and efficiency

Comparison of different algorithms on accuracy and operation time is processed. The YOLOv3 with different batches for training, Faster R-CNN and SSD network are chosen. For the fairness of the comparison, the data set was disrupted. All 45788 images in the dataset are used for training and testing. The training set (24654), verification set (10566) and testing set (10568) are divided according to the same ratio of 7:3:3. All the training and testing tasks of the comparison are also performed on the same workstation mentioned above. In this case, the average precision (AP) value/ F1 score and operation time of each network are listed in Table 2.

It can be concluded that Faster R-CNN has relatively high accuracy of AP, while YOLOv3 has little difference in the recognition accuracy of each type of disease. However, on inference speed, the YOLOv3 is 10 times faster than Faster R-CNN, and is only 70% of SSD. It shows the potential of YOLOv3 on similar accuracy and better efficiency of PD detection and classification. Besides, YOLOv3 with batch64 performs better than bath96.

As calculation, the amount of image data can reach approximately 300pic per kilometre (1 mile = 1.609344 km). It will take 0.0347s to process each image, if with the storage of about 0.6MB. Therefore, it would take 16.75s(1.609344 km*300pic/km*0.0347s/pic = 16.75s) for the method to detect and classify the distresses of one mile of a road lane.

Different scenarios comparison

The biggest challenge of automated PD detection is to consistently achieve high performance under various complex

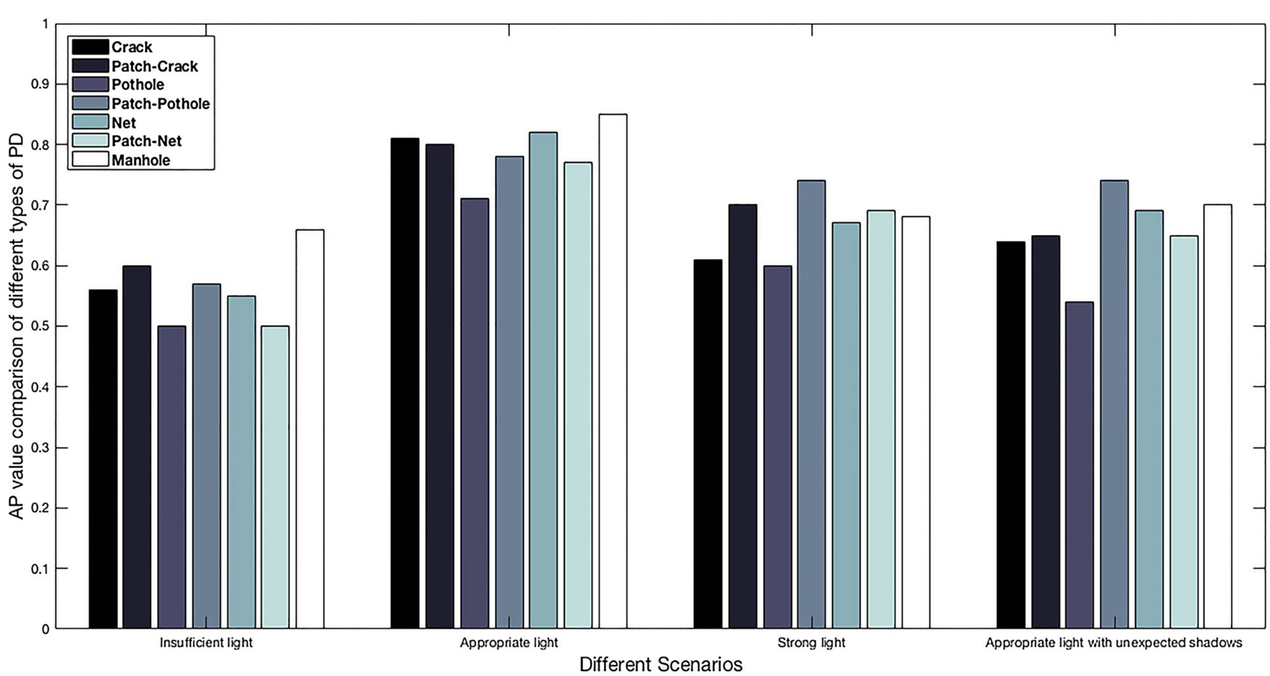


Figure 11. AP value comparison of different types of PD on different image illuminance.

environments. Illumination is a key factor that may cause different environmental conditions. A certain automated algorithm may yield detection results of satisfying accuracy on some particular illumination, while resulting in completely unacceptable error rates on other conditions. Such inconsistent performances may be frequently observed on asphalt surfaces where the textures and roughness levels vary because of the illumination.

The dataset is further divided into 4 groups, which contain images captured on conditions of insufficient light ($0 \leq$ Pixel brightness ≤ 85), appropriate light ($85 <$ Pixel brightness ≤ 170), sufficient light ($170 <$ Pixel brightness ≤ 255) and appropriate light with unexpected shadows (Figure 10). Then F1 score / AP value of different types of PD on different illuminance is calculated and compared (Table 3, Figure 11).

It is obvious that on the condition of appropriate light, all types of PD can be recognized better than on other conditions. Strong light has less impact on object detection process than insufficient light. When main information of pavement is blocked by the shadow, detection becomes more difficult. It is suggested that the image acquisition work be carried out under the condition of appropriate illuminance, while neither at noon with strong illumination nor at night with severely insufficient illumination.

Conclusions

We propose a new technique to detect and classify PD based on the YOLO network. Comparing with traditional PD detection methods, the proposed deep learning method saves more labour cost and is fast to operate. The method is able to meet the increasing demand from road management for distress detection and to utilize the abundant information from pavement images.

The pavement images are collected by high resolution industrial cameras installed on the vehicles, and the distress dataset is generated through the unified labelling process. The dataset is divided into 3 parts for training, cross validation, and testing, respectively.

Comprehensive detection accuracy of 7 types of distress reaches 73.64%. This study shows that PD detection and classification based on YOLO network is feasible. In view of the overall situation, with the increasing size of samples for training, the performance of the network improves.

In terms of the distress detection of each type, higher detection accuracy is normally associated with larger sample size. In addition, if the data size of a certain PD type increases, the detection accuracy also improves. Although the quality and quantity of training data sets are the key to improve the performance of the algorithm, higher cost in data collection and labelling may be involved. YOLOv3 shows the potential on accuracy and efficiency of PD detection and classification. It has slightly difference in the recognition accuracy of each type of disease from Faster R-CNN, while on inference speed, the YOLOv3 (0.0347s/pic) is 9 times faster than Faster R-CNN and is only 70% of SSD.

Looking into the scenarios with various illumination, under the condition of appropriate light, all types of PD can be recognized better than with strong light and insufficient light. When main information of pavement is blocked by the shadow, detection becomes even more difficult.

For future work, more convolution layer structures in the deep network are suggested for further exploration. The specific trend of accuracy improvement is also worth future work to test. Ensemble Learning, as a boosting method, is widely used in classification and regression tasks, which can also be used to improve the overall performance of PD detection and classification

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Author contribution statement

The authors confirm contribution to the paper as follows: study conception and design: Pan. Author, Du. Author; data collection: Pan. Author, Deng. Author; analysis and interpretation of results: Pan. Author, Deng. Author, Shen. Author, Kang. Author; draft manuscript preparation: Du. Author, Pan. Author. Shen. Author. All authors reviewed the results and approved the final version of the manuscript.

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